OLEKSIY OSTAPENKO

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EDUCATION

University of Montreal / Mila, Montreal, Canada *2019 – 2024*
Ph.D. in Computer Science, supervised by Prof. Laurent Charlin. Research focus: continual learning and modular deep learning. Ph.D. thesis: "Towards maintainable machine learning development through continual and modular learning"

Humboldt-Universität zu Berlin, Berlin, Germany *2015 – 2019*
M.Sc. in Information Systems.

Cooperative State University Baden-Württemberg, Mannheim, Germany *2012 – 2015*
B.Sc. in Information Systems

WORK EXPERIENCE

Senior Research Scientist *Nov 2024 – Present*
ServiceNow AI Research (Foundation Models Lab), Montreal, Canada
Research on pre- and post-training of large language models, with a focus on data curation, efficient inference via transformer-to-SSM hybrid distillation (Apriel-H1, Super Apriel), and high-throughput discrete diffusion models. Contributor to Fast-LLM, ServiceNow's open-source distributed LLM pre-training library.

Visiting Researcher *Mar 2023 – May 2024*
Microsoft Research, Montreal, Canada
Research on instruction fine-tuning of LLMs and mixture-of-experts models built from parameter-efficient adapters (e.g., LoRA); led to ICML and COLM publications.

Visiting Researcher *Feb 2022 – Oct 2022*
ServiceNow, Montreal, Canada
Explored continual learning with modular neural architectures and adapter-based networks.

Associate Researcher *Apr 2019 – Sep 2019*
SAP SE, Berlin, Germany
Worked on continual learning using generative replay with GANs.

Working Student, Deep Learning Research Team *Feb 2018 – Mar 2019*
SAP SE, Berlin, Germany
Researched continual (lifelong) learning, few-shot learning, and efficient deep learning. Co-authored papers accepted at CVPR, NeurIPS (Workshop), ICASSP, and WACV.

Working Student, Applied Deep Learning Team *Nov 2017 – Feb 2018*
Hella Aglaia GmbH, Berlin, Germany
Distributed model training on Kubernetes and synthetic data generation for autonomous driving.

Corporate Student *Oct 2012 – Sep 2015*
E.ON SE, Germany / Spain
Database development, data warehousing, and project management across business and development teams.

PUBLICATIONS

Super Apriel: One Checkpoint, Many Speeds
arXiv:2604.19877 *2026*

Oleksiy Ostapenko, Raymond Lin, Torsten Scholak, Akshay Kalkunte, Alireza Mousavi Hosseini, Aman Tiwari, Denis Kocetkov, Joel Lamy-Poirier, Kelechi Ogueji, Nanda Krishna, Pulkit Pattnaik, Rafael Pardinas, Sathwik Tejaswi Madhusudhan, Shruthan Radhakrishna, Srinivas Sunkara, Valérie Bécaert

Breaking the Bottleneck: High-Throughput Diffusion LMs with Mamba Backbone
ICML 2026 *2026*
Vaibhav Singh, Oleksiy Ostapenko, Pierre-André Noël, Torsten Scholak

Apriel-H1: Towards Efficient Enterprise Reasoning Models
arXiv:2511.02651 *2025*
Oleksiy Ostapenko, Luke Kumar, Raymond Li, Denis Kocetkov, Joel Lamy-Poirier, Shruthan Radhakrishna, Soham Parikh, Shambhavi Mishra, Sebastien Paquet, Srinivas Sunkara, Valérie Bécaert, Sathwik Tejaswi Madhusudhan, Torsten Scholak

Using Scaling Laws for Data Source Utility Estimation in Domain-Specific Pre-Training
Multilingual Data Quality Signals Workshop, COLM 2025 *2025*
Oleksiy Ostapenko, Charles Guille-Escuret, Luke Kumar, Max Tian, Denis Kocetkov, Gopeshh Subbaraj, Raymond Li, Joel Lamy-Poirier, Sebastien Paquet, Torsten Scholak

Apriel-Nemotron-15B-Thinker Technical Report
arXiv:2508.10948 *2025*
SLAM Lab, ServiceNow

Exploring Sparse Adapters for Scalable Merging of Parameter Efficient Experts
COLM 2025 *2025*
Samin Yeasar Arnob, Zhan Su, Minseon Kim, Oleksiy Ostapenko, Riyasat Ohib, Esra' Saleh, Doina Precup, Lucas Caccia, Alessandro Sordoni

Towards Modular LMs by Building and Reusing a Library of LoRA Adapters
ICML 2024 *2024*
Oleksiy Ostapenko, Zhan Su, Edoardo Ponti, Laurent Charlin, Nicolas Le Roux, Lucas Caccia, Alessandro Sordoni

Guiding Language Model Reasoning with Planning Tokens
COLM 2024 *2024*
Xinyi Wang, Lucas Caccia, Oleksiy Ostapenko, Xingdi Yuan, Alessandro Sordoni

A Case Study of Instruction Tuning with Mixture of Parameter-Efficient Experts
NeurIPS 2023 Workshop on Instruction Tuning and Instruction Following *2023*
Oleksiy Ostapenko, Lucas Caccia, Zhan Su, Nicolas Le Roux, Laurent Charlin, Alessandro Sordoni

Challenging Common Assumptions about Catastrophic Forgetting in Continual Learning
CoLLAs 2023 *2023*
Timothée Lesort, Oleksiy Ostapenko, Pau Rodriguez, Diganta Misra, Md Rifat Arefin, Laurent Charlin, Irina Rish

From IID to the Independent Mechanisms Assumption in Continual Learning
Continual Causality Bridge at AAAI 2023 (Oral) *2023*
Oleksiy Ostapenko, Pau Rodriguez, Alexandre Lacoste, Laurent Charlin

Continual Learning with Foundation Models: An Empirical Study of Latent Replay
CoLLAs 2022 (Oral) and CVPR 2022 Workshop on Continual Learning (Oral) *2022*
Oleksiy Ostapenko, Timothée Lesort, Pau Rodriguez, Md Rifat Arefin, Arthur Douillard, Irina Rish, Laurent Charlin

Continual Learning via Local Module Composition
NeurIPS 2021 *2021*
Oleksiy Ostapenko, Pau Rodriguez, Massimo Caccia, Laurent Charlin

Online Fast Adaptation and Knowledge Accumulation: A New Approach to Continual Learning
NeurIPS 2020 *2020*
Massimo Caccia, Pau Rodriguez, Oleksiy Ostapenko, Fabrice Normandin, Min Lin, Lucas Caccia, Issam Laradji, Irina Rish, Alexandre Lacoste, David Vazquez, Laurent Charlin

Learning to Remember: A Synaptic Plasticity Driven Framework for Continual Learning
CVPR 2019 *2019*
Oleksiy Ostapenko, Mihai Puscas, Tassilo Klein, Patrick Jähnichen, Moin Nabi

Self-Paced Adversarial Training for Multimodal Few-Shot Learning
IEEE Winter Conference on Applications of Computer Vision (WACV) 2019 *2019*
Frederik Pahde, Oleksiy Ostapenko, Patrick Jähnichen, Tassilo Klein, Moin Nabi

Prune Your Neurons Blindly: Neural Network Compression through Structured Class-Blind Pruning
ICASSP 2019 *2019*
Abdullah Salama, Oleksiy Ostapenko, Tassilo Klein, Moin Nabi

OTHER ACTIVITIES

Reviewing
Peer reviewer for NeurIPS, ICML, CVPR, TPAMI, and AISTATS etc.

Co-organizer, AI Helps Ukraine Conference *2022*
Organized an AI conference to fundraise for humanitarian aid for Ukraine (aihelpsukraine.cc)

Data Science Game 2017, Paris, France (Jury Award) *2017*
Qualified among the top 20 of 200+ teams and competed in the final; solved a time-series demand prediction task for a music recommendation system

Co-founder, TEDxHUBerlin, Berlin, Germany *2017*
Co-founded a non-profit organization spreading local "ideas worth spreading"

TECHNICAL SKILLS

Programming Languages	Python, Java
Frameworks & Tools	PyTorch, Fast-LLM (distributed LLM training), NumPy, scikit-learn

LANGUAGES

Ukrainian & Russian (Native), **German** (Near Native), **English & Spanish** (Fully Professional), **French** (B1)